



**A.Y:2025-2026**

**ASSIGNMENT QUESTION PAPER FOR INTERNAL EXAMINATIONS**

**Program : B.TECH**

**Date :19/11/2025**

**Branch : CSM/ECE/CSE(ICB)/CSE(CS)/CSE(AI&DS)/CSE(AI)/R&AI/CSIT**

**Year : I**

**SEM: I**

**MID: II**

**Regulation:MRV-25**

**Subject name: Fundamentals Of Quantum Computing    Subject Code: 25FQM101**

**Max Marks: 10**

**Blooms level:**

|            |           |         |           |          |           |
|------------|-----------|---------|-----------|----------|-----------|
| Remember   | <b>L1</b> | Apply   | <b>L3</b> | Evaluate | <b>L5</b> |
| Understand | <b>L2</b> | Analyze | <b>L4</b> | Create   | <b>L6</b> |

| S.NO     | ANSWER ALL QUESTIONS                                                                                                                                                              | Course Outcomes |                                   |     | Blooms level | Marks |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----------------------------------|-----|--------------|-------|
|          | PART-A( 5Q X 2=10 MARKS)                                                                                                                                                          | CO              | PO                                | PSO |              |       |
| <b>1</b> | Write the matrix representations of Hadmard (H) gate                                                                                                                              | CO3             | 1,2,3,<br>4,5,6,<br>7,8,10<br>,12 | 1,2 | L2           | 2     |
| <b>2</b> | Find the final qubit state for the circuit —[ Z]—[ X]—<br>Using inputs $ 0\rangle$ and $ 1\rangle$                                                                                | CO4             | 1,2,3,<br>4,5,6,<br>7,8,10<br>,12 | 1,2 | L1           | 2     |
| <b>3</b> | Find the final state of the following circuit:<br><br><div style="text-align: center;"> <math> 0\rangle \text{ — [ H ] —}</math><br/> <math> 1\rangle \text{ —————}</math> </div> | CO4             | 1,2,3,<br>4,5,6,<br>7,8,10<br>,12 | 1,2 | L1           | 2     |
| <b>4</b> | Explain the major applications of quantum computing in molecular modelling and new material design                                                                                | CO5             | 1,2,3,<br>4,5,6,<br>7,8,10<br>,12 | 1,2 | L5           | 2     |
| <b>5</b> | Discuss the applications of quantum computing in deep learning                                                                                                                    | CO5             | 1,2,3,<br>4,5,6,<br>7,8,10<br>,12 | 1,2 | L6           | 2     |

**Submission Last Date: 29/11/2025**